

Shodan (Sentient Hyper-Optimized Data Access Network) is a **search engine** that scans the internet for connected devices and services.

Website: <https://www.shodan.io>

It allows filtering based on ports, services, countries, hostnames, banners, etc.



Shodan Search Command Format

<search term> filter1:value filter2:value



Common Shodan Search Filters

Filter	Description	Example
ip:	Search a specific IP address	ip:8.8.8.8
hostname:	Match specific hostname or domain	hostname:example.com
port:	Devices running specific ports	port:22, port:80
country:	Country code (ISO 2-letter)	country:IN, country:US
city:	Devices in a specific city	city:Mumbai
org:	Internet service provider or organization	org:Reliance Communications
asn:	Autonomous System Number	asn:AS15169
os:	Operating system info	os:Windows, os:Linux
product:	Product name (e.g., Apache, nginx, SSH)	product:Apache, product:nginx
version:	Version of product or service	product:Apache version:2.4.7
before:	Devices found before a specific date	before:2022-01-01
after:	Devices found after a specific date	after:2023-01-01
ssl:	Devices with SSL cert fields	ssl:expired:true, ssl.cert.issuer.cn:"Google"
vuln:	Search for specific CVEs (vulnerabilities)	vuln:CVE-2021-26855
title:	Title of web application	title:"webcamXP"
net:	CIDR range search	net:192.168.1.0/24



Useful Example Searches

Goal	Shodan Query Example
Find open webcams	product:"webcamXP" or inurl:view/index.shtml
Search for SSH services in India	port:22 country:IN
Find Apache servers in the US	product:Apache country:US
Look for exposed MySQL	port:3306
Find devices with expired SSL	ssl:expired:true

Find devices vulnerable to CVE	vuln:CVE-2021-26855
View Microsoft Exchange Servers	product:"Microsoft Exchange"
Find devices running Windows	os:"Windows" port:445
CCTV/DVR search	title:"DVR login"
Industrial control systems	port:502 modbus



Bonus Tips

- Combine filters:

port:22 os:"Linux" country:IN org:"BSNL"

- Use **quotes** for multi-word filters:

product:"Apache Tomcat"



Legal Note:

- **Use Shodan only for authorized ethical hacking or OSINT research.**
- Scanning or attacking exposed devices without permission is **illegal**.